

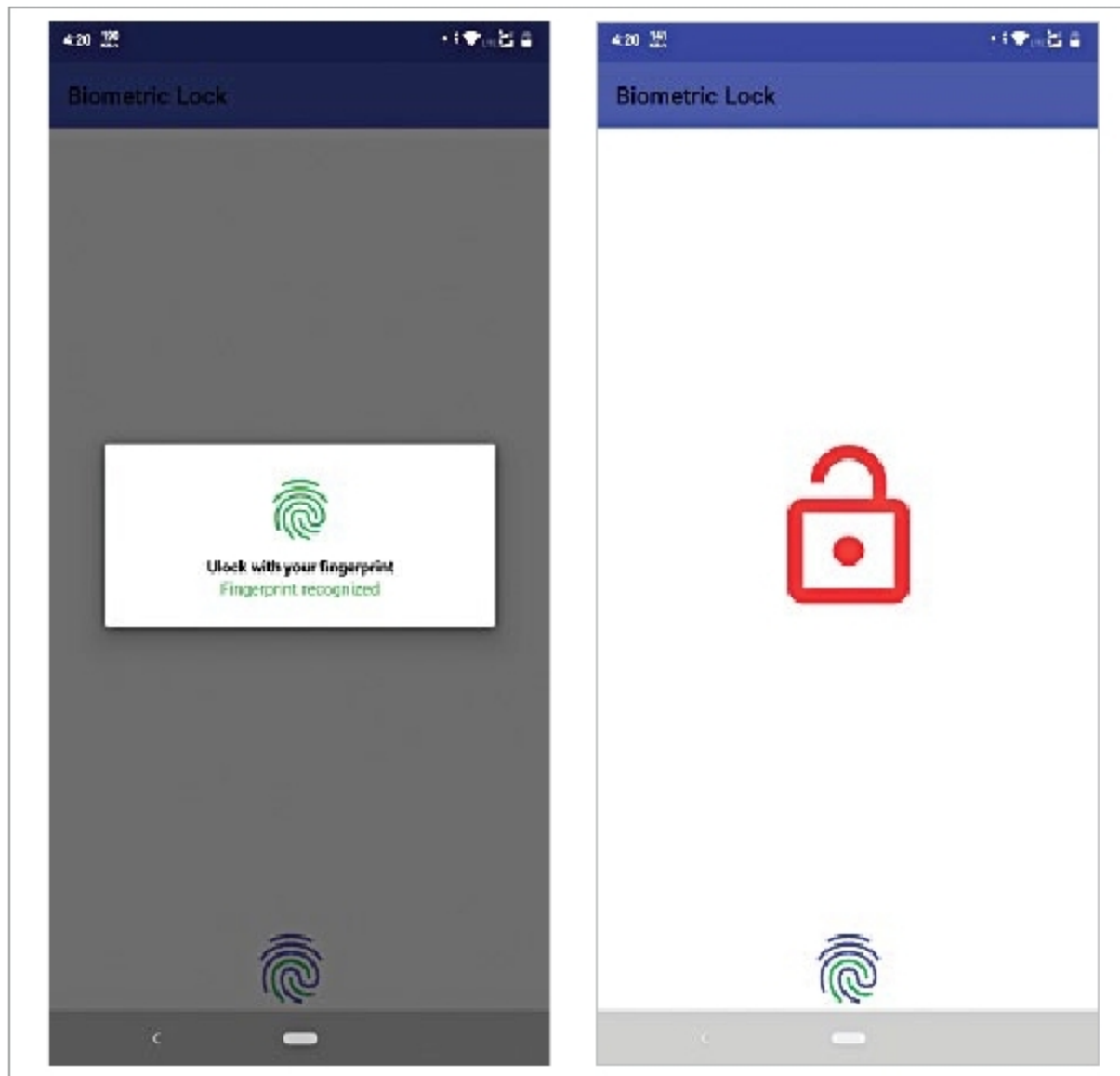


Fig. 11: App prompting you to unlock using your finger

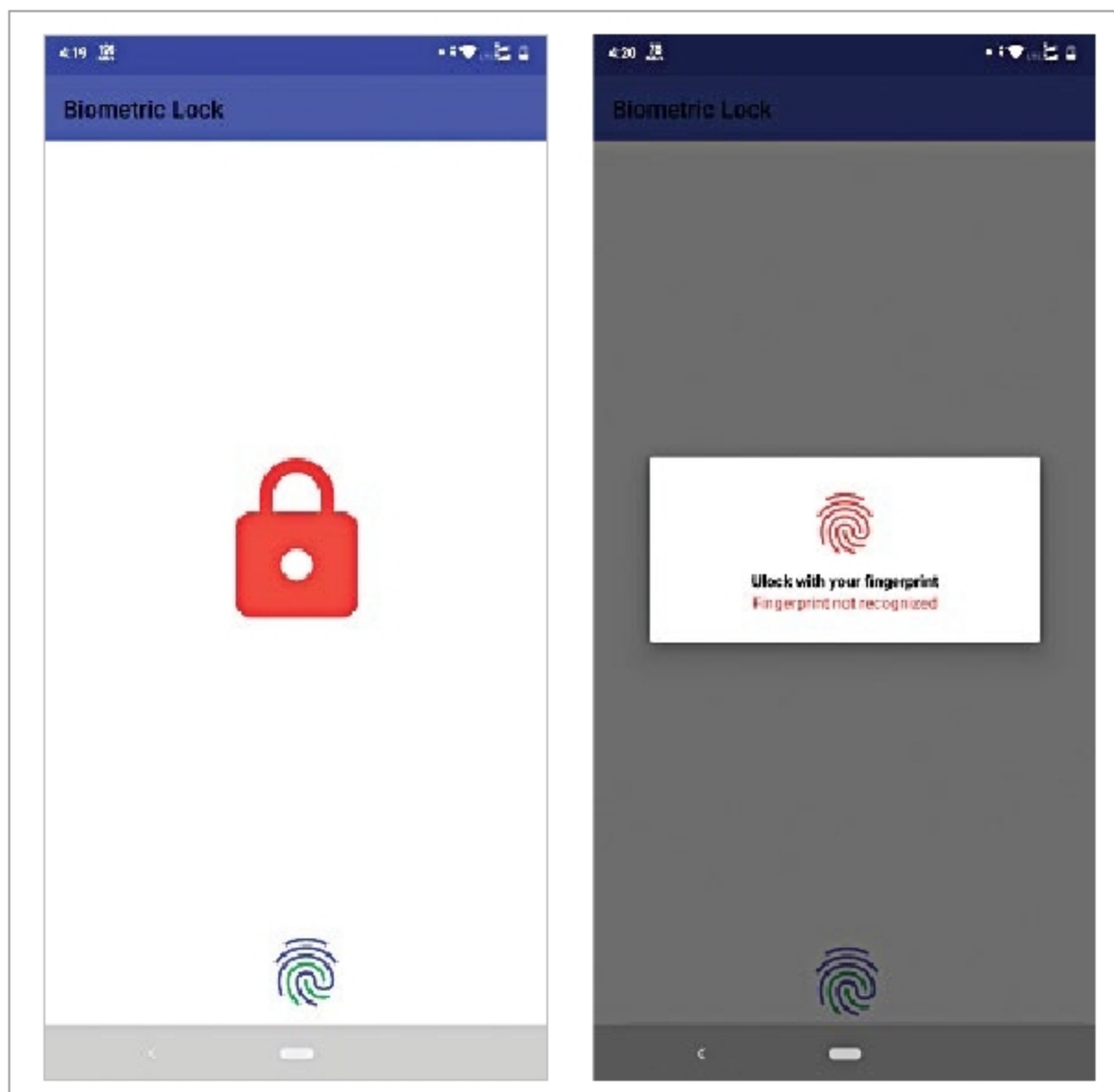


Fig. 12: Fingerprint does not match prompt

Next, join the code blocks as shown in Fig. 4. Save the code as, say, Biometric_Lock. Then, export Biometric_Lock.apk file to your PC or laptop. Transfer .apk file from the PC to the smartphone. Install .apk file on the smartphone. With this, software development including Arduino coding and app development is complete.

Connect components to Arduino board as given in Table 2 and shown in Fig. 5.

Interfacing the circuit with the lock

Follow the steps given in Figs 6 through 9 to make the biometric locking system.

Step 1. Get a suitable lock from the hardware store.

Step 2. Open the cover of the lock and you will be able to see the lever, as shown in Fig. 7.

Step 3. Attach one of the servo motor fins to lever of the lock, as shown in Fig. 8.

Now, put back the cover of the lock to close it. Extend three wires of the servo from the lock to the circuit. Make sure all wires are connected to appropriate points in the circuit.

Testing

Power on the circuit with 5V-6V battery. Turn on Bluetooth of the phone and pair with Bluetooth HC-05 in the circuit. After successful connection with Bluetooth HC-05, open Biometric_Lock app on the smartphone. You will get an app layout with Bluetooth and fingerprint icons. Touching Bluetooth icon changes into locked icon (Fig. 10).

After touching the fingerprint icon, you will get a dialogue box on the screen asking you to unlock using your fingerprint (Fig. 11). Touch the fingerprint sensor, which is normally at the back of the smartphone. If it is the correct finger, the servo will rotate to turn the lock on, and at the same time the locked icon will change to unlocked icon.

If fingerprint does not match, the lock will remain in locked position. The dialogue box will give error message as "Fingerprint not recognised" (Fig. 12). **EFY**

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